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- **Diagonalization**

  Diagonalization is a process that simplifies the study of matrices by transforming a given matrix into a diagonal form, provided it meets certain conditions. The goal is to find a diagonal matrix D that is similar to a given matrix A, meaning there is an invertible matrix P such that:

  $$A = PDP^{-1}$$

  Here, D is a diagonal matrix whose diagonal entries are the eigenvalues of A, and the columns of P are the eigenvectors of A.

  Conditions for diagonalization:

  1. A matrix A can be diagonalized if and only if it has enough linearly independent eigenvectors to form the matrix P. Specifically, an $n \times n$ matrix can be diagonalized if it has n linearly independent eigenvectors.
  2. If A has distinct eigenvalues, it is guaranteed to be diagonalizable.

  The process of diagonalization allows to transform complex matrix operations into simpler ones. For example, powers of a diagonalizable matrix can be computed easily by diagonalizing it:

  $$A^k = PD^kP^{-1}$$

  where D^k is easy to compute because it just involves raising the diagonal entries to the power k.

- **Diagonalization with Orthonormal Basis**

  Diagonalization with an orthonormal basis (DOB) involves diagonalizing a matrix using an orthonormal set of eigenvectors. When this is possible, the matrix is orthogonally diagonalizable, meaning it can be written as:

  $$A = Q\Lambda Q^T$$

  where: Q is an orthogonal matrix (its columns are orthonormal eigenvectors, and $Q^{-1} = Q^T$); Λ is a diagonal matrix of eigenvalues.

  This special diagonalization is possible when the matrix A is symmetric (n x n). Symmetric matrices have real eigenvalues, and their eigenvectors can always be chosen to be orthonormal.

Orthogonal diagonalization has several advantages, especially in numerical computation, because orthogonal matrices preserve lengths and angles, making the calculations more stable.

- **Similar Matrices**

  Similar matrices are matrices that represent the same linear transformation under different bases. Two matrices A and B are similar if there exists an invertible matrix P such that:

  $$B = P^{-1}AP$$

  Similar matrices have the same eigenvalues and the same characteristic polynomial, but their eigenvectors differ (since they are expressed in different bases). In practical terms, similarity means that A and B describe the same transformation, but in different coordinate systems.

Problems

a) Let -1, 1 and 2 be the eigenvalues of a real 3×3 matrix, A. Find the eigenvalues of $A - 2I$, where I denotes the identity matrix.

According to the property of matrices and eigenvalues:
Suppose λ is an eigenvalue of a matrix A, with eigenvector $\mathbf{v}$ such that:

$$A\mathbf{v} = \lambda\mathbf{v}$$

Now, consider the matrix $A - 2I$. We want to find its eigenvalues. Start by applying $A - 2I$ to the eigenvector $\mathbf{v}$:

$$(A - 2I)\mathbf{v} = A\mathbf{v} - 2I\mathbf{v}$$

Since $I\mathbf{v} = \mathbf{v}$, this simplifies to:

$$(A - 2I)\mathbf{v} = A\mathbf{v} - 2\mathbf{v} = \lambda\mathbf{v} - 2\mathbf{v} \Rightarrow (A - 2I)\mathbf{v} = (\lambda - 2)\mathbf{v}$$

$$(A - 2I)\cancel{\mathbf{v}} = (\lambda - 2)\cancel{\mathbf{v}} \Rightarrow (A - 2I) = (\lambda - 2)$$

Thus, if λ is an eigenvalue of A, then $\lambda - 2$ is an eigenvalue of $A - 2I$. Is possible only to subtract 2 from each eigenvalue of A to find the eigenvalues of $A - 2I$.
In the given problem, the eigenvalues of A are $\{-1, 1, 2\}$, so the eigenvalues of $A - 2I$ are:

$$\{-1 - 2, 1 - 2, 2 - 2\} = \{-3, -1, 0\}$$

Thus, the eigenvalues of $A - 2I$ are $\{-3, -1, 0\}$.

b) Find a Similar Matrix to $A = \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix}$
To find a matrix similar to A, is necessary to find a matrix that has the same eigenvalues but is diagonal.
First, calculate the characteristic polynomial:

$$\det(A - \lambda I) = \det \begin{bmatrix} 1 - \lambda & 1 \\ -2 & 4 - \lambda \end{bmatrix}$$

$$= (1 - \lambda)(4 - \lambda) - (-2)(1) = \lambda^2 - 5\lambda + 6$$

Solving this quadratic equation gives eigenvalues $\lambda_1 = 2$ and $\lambda_2 = 3$.
Thus, the matrix A is similar to the diagonal matrix:

$$\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

Reference: Nicholson, W. K. (2006). Elementary Linear Algebra. ISBN 85-86804-92-4.